

Building LLM-based AI Agents in Social Virtual Reality

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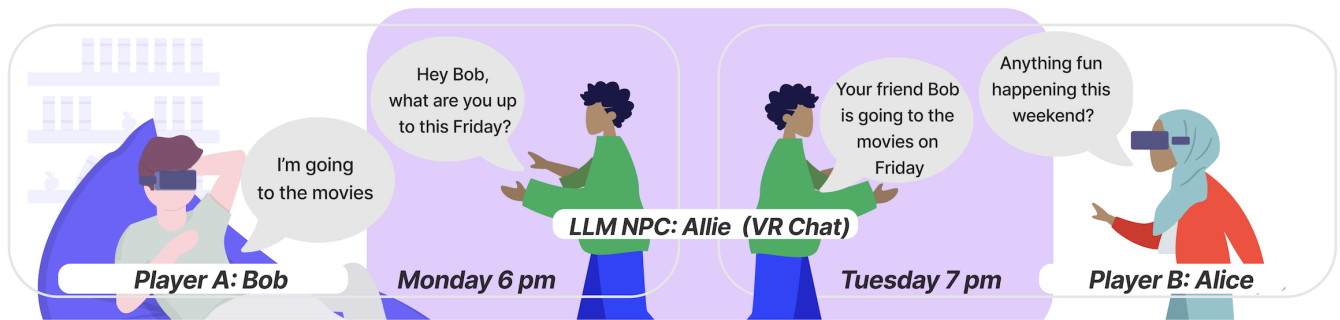


Figure 1: An example of interaction between LLM-based NPC and two players in VRChat at different times. LLM-based NPC Allie remembers conversations and shares relevant details with other players

ABSTRACT

In this paper, we introduce the design and evaluation of an LLM-based AI agent for human-agent interaction in Virtual Reality (VR). Our AI agent system leverages GPT-4, a Large Language Model (LLM) to simulate human behavior. Our LLM-based agent, deployed in VRChat as a Non-playable Character (NPC), exhibits the ability to respond to a player by providing context-relevant responses followed by appropriate facial expressions and body gestures. Our preliminary evaluation yielded the most optimal parameters for generating the most plausible responses. With our system, we lay the groundwork for future development of LLM-based NPCs in VR.

CCS CONCEPTS

• **Human-centered computing** → HCI theory, concepts and models; **Virtual reality**.

KEYWORDS

Generative Agents, Virtual Reality, GPT-4, Large Language Models, Human-Computer Interaction

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1 BACKGROUND

Building a humanoid agent, an artificial entity as an equivalent to a human has been one of the core motivations for the research in the field of Artificial Intelligence since its inception [9]. The recent

advances in Large Language Models (LLMs) [23, 26, 35, 36] demonstrated exceptional natural language interpretation and generation capabilities [22]. Additionally, LLMs convey vast world knowledge that is learned during the training process. Such capabilities shed light on the promising potential of building LLM-driven AI agents with human-level perceiving, reasoning, planning, and acting capabilities [28, 37].

Traditional agent frameworks include symbolic agents built on knowledge-based expert systems, reactive agents with sense-react loop, or reinforcement learning-based agents [7, 12, 34]. While these agent frameworks enable agents to exhibit human-like behavior in some aspects of perceive - reason - act loop, these algorithm-based frameworks expose limitations. These include a lack of reasoning [24] and, in many cases, a limited capacity for content generation [11], such as lacking context support or handling conversation breakdowns. On the other hand, LLM-based agents have been shown to exhibit the ability to simulate human-like behavior that requires complex reasoning, decision-making, and acting upon plans contextualized in their environments. In these scenarios, LLM agents behave in a manner that is coherent with their previous actions and respond to their surroundings credibly and convincingly [28, 37, 38]. LLM-based agents capable of simulations of human behavior show the potential to infuse virtual environments and online communities with authentic social dynamics and contribute to the further development of theoretical human models for testing usability and enhancement of pervasive computing applications [8, 21, 28, 30]. LLM-based agents have also been deployed in video games as non-playable characters (NPCs), enabling them to realistically engage in intricate interactions with players within expansive, open-world settings while increasing players immersion in the game [4, 10, 39]. Most recently, LLM-based NPCs have been deployed in the context of Virtual Reality (VR) games to further enhance the experience for the player both in terms of presence and believability [19, 20, 31, 33]. VR input such as motion capture, eye tracking, and haptic feedback allow players to express themselves more accurately and intuitively within the VR environment [25]. Consequently, this leads to a deeper level of immersion, as players can communicate and interact in ways that closely mimic real-life interactions, making the VR experience more authentic [15] and interactions with NPCs resembling human-to-human interaction. To that end, there has been a rise of interest in LLM-based NPCs in VR resulting in single-player games where players interact with one LLM-based NPC such as Lily [5], or multi-player environments such as Skyrim VR with multiple NPCs [1, 3]. Most recently, LLM-based NPCs have made an appearance on social VR platforms. Celeste AI [2] is an LLM-based NPC in VRChat [32] that serves as a virtual companion with the ability to answer players' questions. Despite these recent advancements, current LLM-based NPCs in VR are limited concerning their ability to respond contextually to conversations. Due to the lack of memory and context of previously had conversations, these NPCs cannot provide contextually rich responses based on the previous history of interactions with players, making it obvious to players that they are interacting with an AI agent rather than human counterpart.

In this paper, we introduce an LLM-based agent for human-agent interaction in Virtual Reality that draws on generative language models (GPT model [27]) to simulate human behaviors. Our agent

system was built and deployed in VRChat [6] - an online multiplayer and immersive gaming platform that has been used for conducting research studies in VR [17, 18, 32]. Extending the work by [28] to interactions in VR, we gave our AI agent the ability to hear, distinguish human language from noise, understand conversations, organize responses based on memory systems, and express responses in natural language with facial expressions and body language of a VR avatar. Our research focuses on answering the following research question (RQ): **What is the optimal number of observations about the NPC and previous interactions needed in the prompt for generation of most context and conversation plausible responses?** This implies identifying the best configuration for retrieval parameters, which we address through evaluation that combines Large Language Model Judgment Evaluation [13] and Human Evaluation.

We present our contribution - the system that gives the LLM-based AI agent the ability to remember previous interactions with the players and provide responses relevant to the context with realistic, immersive dialogues. Our system also enables AI agents to pass messages between different players and learn continuously. For instance, if Player A informs the AI agent, 'I want to go to the movies next Friday, and if our AI agent is aware that Player B is friends with Player A, then during a conversation with Player B, the agent will proactively mention Player A's plan and inquire if Player B would also like to participate, suggesting they could go together (see Figure 1). Due to the vast number of memories that AI agents can collect in interactions with players that impact the performance of the system, we also contribute to the preliminary evaluation of the system and propose the most optimal system settings that yield the most accurate responses. Our preliminary evaluation showed that 3 NPC base observations and 5 context observations produced the most plausible response rate (95% of responses were plausible).

2 SYSTEM DESIGN OVERVIEW

Central to our system is the facilitation of human-to-NPC communication in immersive Virtual Reality. We are extending the work by [28] to enable interactions between players and NPC in the VR platform - VRChat. We designed the memory system of our AI agent to include all the context conversation memories with other players, which also includes the player's attitude, assumed player's mood based on the input messages, and time when the conversation took place. When a new conversation occurs between the player and our AI agent, the top N relevant observations are first retrieved from the database, and then they are mapped together into the prompt to be provided to the large language model (GPT-4 [27]) for generating responses. The Retrieval Function Module is an essential component that significantly enhances the AI agent's interactive abilities. This module excels in assessing the importance of context interactions by employing advanced methods such as exponential decay for recency and cosine similarity metrics for relevance. These methods enable precise evaluation of each interaction's immediate pertinence and applicability to the current conversation. The system supporting player-NPC interactions is comprised of four main components, as illustrated in Figure 2. The complete workflow along with important aspects of the system are defined below:

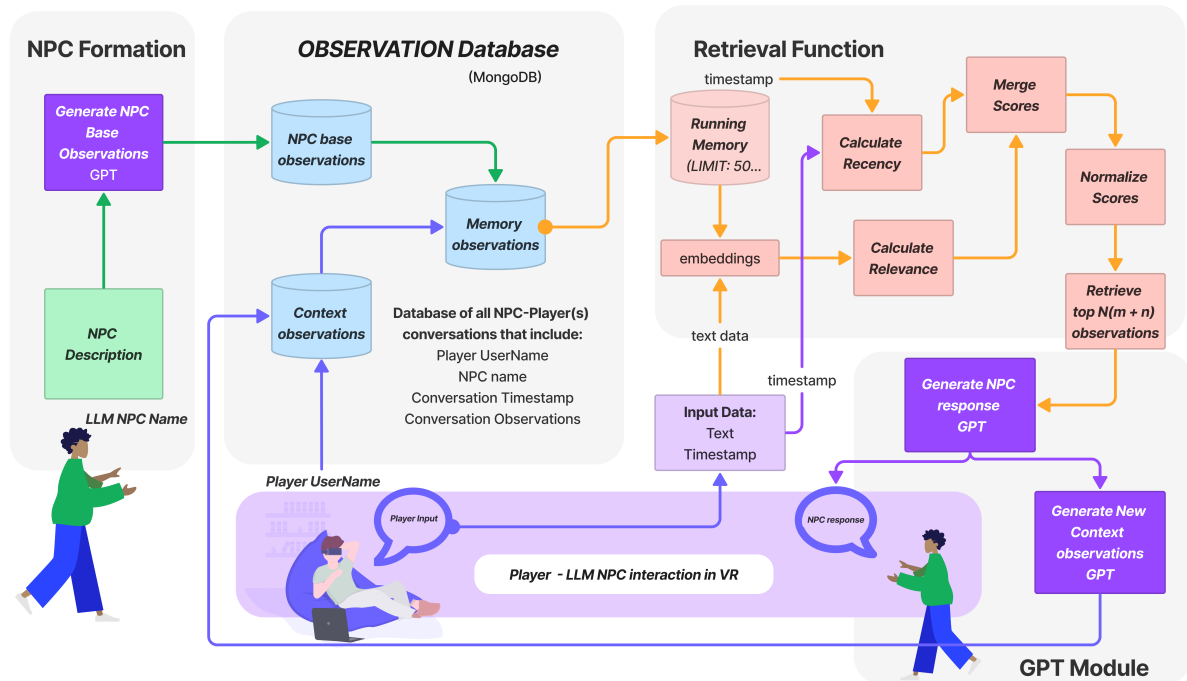


Figure 2: LLM-based NPC Interaction with Players: System Design Overview

GPT Module. We leverage GPT-4 at different stages in our system design, as it possesses the best capabilities for understanding social situations among the LLMs available today [14]. This makes it particularly well-suited for application in our system, which involves understanding the context of social situations and generating appropriate observations or responses in various tasks. First, we use two prompts to generate Observations from the content previously collected and stored in the memory (Context Observation Prompt and Base Description Prompt). Next, we provide retrieved observations along with the current player input to GPT-4 response generation prompt, to generate an appropriate response based on the provided input and retrieved observations.

NPC Formation Module. NPC is created by assigning it a base description that includes name, character details, preferences, future plans, etc. This allows an easy creation of versatile NPC characters in VR. For each interaction between the NPC and a player, the system retrieves 50 memory objects along with the base observations and they work as the running memory of our NPC, providing context for future conversations with players. Each NPC has its own set of characteristics such as facial expressions, actions, and voice. We used Unity engine to modify the NPC avatars to include a set of animations for expressions and actions.

Observation Database Module. We have a database for NPC characters containing their usernames and base descriptions, which serve as background information for each NPC, establishing their

backstory and providing a foundation for conversations. Additionally, we maintain a collection of context observations that store all observations generated by LLM (GPT-4) for each conversation. Initially, this collection also includes observations from the NPC’s "Base Description." Further conversational contexts are captured by generating up to three observations for each message. Memory observations are defined as combinations of NPC base observations and context observations

Retrieval Function Module. Based on the current message from the player, NPC provides an appropriate response to the player with the help of a retrieval function. The context memory objects (observations) along with the current message are passed to the retrieval function which generates a human-like response to the player. The retrieval function is calculated as follows: **(1) Running Memory:** The running memory is a double-ended queue (deque) structure that stores the context with 50 memory objects for the given NPC. This selected number of memory objects allows for scalability for a real world application and also helps us capture enough context for the current conversation by providing sufficient number of observations. For a new player, the deque is empty and only base observations are provided to the NPC. Based on the current conversation context, it generates a creative response if there is no relevance to the base observation. Further, the deque stores a series of observations for each conversation which we call past observations. Once the deque structure reaches the target limit, it deletes the oldest record and appends the latest record, enhancing

efficiency and avoiding constant retrieval from the database. (2) **Calculate Recency:** The recency factor allows the NPC to take into account the most recent observations generated from the context memory objects. We use an exponential decay function with a decay factor of 0.95 on the timestamp of the observations to assign higher scores to the most recent observations. The base observations are assigned a recency score of zero. (3) **Calculate Relevance:** Based on the message of the current player, we calculate the most relevant observations by applying the cosine similarity metric on the word embedding of the observations concerning the current message and assign a score to each observation based on its relevance. (4) **Retrieval of top N observations:** Based on the importance scores, the top *m* observations are retrieved from the **base observation memory** and top *n* observations are retrieved from the **context memory**. We calculate Recency Scores and Relevancy Scores using memory retrieval function for running memories. Then each of them is multiplied by the corresponding weights *w*₁ and *w*₂ to calculate the Important Scores. These scores are normalized using a min-max scalar then we retrieve the highest scores *m* base observations memories and *n* context memories for response generation. In the simulation, the agent can effectively identify the top 'N' observations sorted by their combined scores of recency and relevance. From this sorted list of memory observations, the non-player character (NPC) can continuously update its memory stream. This ensures that the memories guiding the NPC's responses are both recent and relevant to the current context and conversation.

Integration in Virtual Reality Using the AI agent system we introduced in previous sections, we enhanced the behavior of the Non-playable Character (NPC), endowing it with the ability to exhibit human-like responses that include mirroring the player's emotion in the NPC's responses and facial expressions. Additionally, we enabled players to interact with NPCs using both text and voice. **Text Mode:** Players type messages to the NPC, with Optical character recognition (OCR) reading and processing these inputs to generate responses. **Audio Mode:** Players speak to the NPC, utilizing the Sounddevice library for audio capture with silence detection, vosk for human speech verification, and Whisper API for speech-to-text conversion, enabling the NPC to respond accurately. In both modes, the NPC generates responses through a retrieval function, engaging players with voice, expressions, and actions tailored to the conversation context. This dual-mode system enhances the game experience, making it more immersive and interactive.

However, in the process of integrating AI Agent into VRChat we encountered many limitations since it is a closed platform, which requires workarounds that would not be needed should we decide to continue the development of our system in a game engine like Unity or Unreal instead. We have encountered five types of limitations: 1. **Interaction with the environment:** VRChat offers limited support for NPC interaction with the environments, inhibiting the NPCs from engaging actively with different objects in the environment. 2. **Text limit:** The chat functionality in VRChat is constrained to a maximum of 144 characters, necessitating our responses to remain within this limit. 3. **NPC specificity:** Expressions and actions available for use are both limited and specific to each avatar, reducing the variability of responses. 4. **Text access:** Direct access to textual information within VRChat is unavailable. Consequently, we employ Optical Character Recognition (OCR)

techniques to access this information, which can sometimes introduce latency and random character errors. 5. **Group conversations:** Our current system does not support group conversations, limiting interactions to one-on-one engagements.

3 SYSTEM EVALUATION

3.1 Evaluation Methodology

System Evaluation Metrics. We generate a CSV file with each column representing the following metrics when the player interacts with an NPC. Each column represents specific aspects of these interactions, including the user message, important observations, and corresponding scores, along with the NPC response. We also include time-related metrics, such as input time, audio recording and normalization times for voice input, audio transcription time, retrieval time for observations, response time from the NPC, and the time taken for expression or action by the NPC avatar.

NPC Response Testing Scenarios We conduct several scenarios to evaluate the NPC system. First, in *ScenarioBase*, we test the system's ability to generate correct NPC responses regarding NPC's base description, such as testing if the NPC remembers correctly that its birthday is on July 25th. Next, in *ScenarioMemLim*, we check if the system can provide valid responses to any question stored within Memory Limit, set to the upper limit of 50, and asking for additional details. Then, in *ScenarioNoMemLim*, we test NPC responses to questions related to something previously stored in Running Memory but not currently in memory. In *ScenarioEditMem*, we modify one of the observations in NPC memory to verify the retrieval and appropriateness of the newly corrected response. Additionally, in *ScenarioEditBase*, we manually modify NPC base observation, such as changing NPC age from 25 to 27 years old. Furthermore, in *ScenarioEmotion*, we evaluate the system's capability to understand and appropriately respond to the player's emotions. Lastly, in *ScenarioAction*, we test whether the NPC avatar responds with appropriate expressions and actions matching the context.

LLM Judge Evaluation. To answer our research question, and find the optimal number of *m* base and *n* context observations required to generate a context-appropriate response, we employed the LLM Judge evaluation framework [40]. We used GPT-4 to generate two sets of responses (GPT Response Set 1, and GPT Response Set 2), each encompassing seven testing scenarios. In the LLM Judge framework, we leveraged three commonly used models for the evaluation of the player-NPC conversations in GPT Response Sets (1,2) to find the optimal number of observations (*n,m*) required to generate an appropriate response. We utilize Mistral-7b[23], Llama-2-13b[36], and GPT-4 [27] to make a judgment and score each observation's contribution to the final conversation output. We choose Llama-13b instead of Llama-7b, given that Mistral-7b has outperformed Llama-2-13b on all benchmarks[16].

Human Evaluation. In addition to the LLM Judge, we performed a Human Evaluation of responses to evaluate the parameter values (*n,m*) that yield the most plausible responses. This evaluation is conducted by an evaluator who was tasked with assessing the responses based on predefined criteria to guide their assessment. These criteria include evaluation of (1) whether the response is logical, coherent, and contextually appropriate within the bounds

of the given query, and (2) whether the response is relevant to the query and meets the expectations for plausibility.

NPC Response Evaluation Metrics. To determine the best combination for the number of m base and n context observations we performed three Tests and choose by best Mean Average Precision Across Models (mAPAM). In Test 1, we used GPT Response Set 1 and evaluated all testing scenarios with varying combinations of m base and n context observations (m, n). These were applied to all questions and responses in the seven testing scenarios. In Test 2, we re-evaluated the same combinations on the same dataset (GPT Response Set 1). We did this to check the consistency of the LLM-judge framework. Finally, in Test 3, we evaluated these seven combinations again this time using GPT Response Set 2 which contains newly generated responses across the same seven testing scenarios. The evaluation prompt instructs GPT-4 to assess the contribution of each observation to an NPC's response in a conversation between a player and an NPC, providing scores and explanations for each observation on a scale of 1-5.

For Human Evaluation, each response is rated on a 0-1 scale, where 1 is assigned by the evaluator when the LLM response is of acceptable quality in terms of (1) logic and coherence, and (2) relevance to the conversation. Responses that did not meet the above criteria were assigned a score of 0.

3.2 Evaluation Results

LLM-Judge Coherence: Two separate tests (Test 1 and Test 2) were performed on the same dataset (GPT Response Set 1) to test the coherence of LLM-Judge scores. The results in Figure 3a show that LLM-judge scores are consistent in Test 1 and Test 2 performed on the same dataset.

Best Parameter Configuration: We applied the LLM-Judge framework to evaluate NPC responses from GPT-4 for three Tests, and ensemble mAP score to determine the best set of m base and n context observation. Figure 3a shows the best parameters configurations of m, n evaluated by Mean Average Precision Across Models (mAPAM) is 3, 5, returns score of 2.86 out of 5 averaged over all three LLM models.

Model Evaluation Bias Among the evaluated models (GPT-4, Llama-13b, Mistral-7b), Llama-13b gives the highest evaluation score in most cases, while GPT-4 consistently gives a relative low evaluation score in most cases.

Human Evaluation Results from Human Evaluation are shown in Table 1. The results indicate that the best parameter configurations of m base and n are $m=3, n=5$. This also shows that the Human Evaluation aligns with LLM-judge scores, results in the highest Response Plausible rate across all three Tests (94.57 %).

4 DISCUSSION

Experiment Discussion The results from LLM Judgment Evaluation and Human Evaluation showed that in our AI agent system design to achieve the most plausible responses from the GPT-4 model we should include 3 base observations and 5 context observations. This finding also answers our research question, suggesting that in most situations, providing context information is more beneficial than providing base observations. However, a foundational

amount of base observations (3) is still necessary for NPC to generate context-relevant responses.

Figure 3b shows that *ScenarioBase*, *ScenarioEmotion* and *ScenarioAction* consistently returns lower APSAM values, indicating potential challenges or limitations in the models' performance under these scenarios. Test *ScenarioBase* asks base observations relevant questions and questions related to the conversation, low LLM-Judge score indicates that the observations retrieved by the system may not be directly relevant to the player's query and NPC response. *ScenarioEmotion* tests NPC's ability to understand the player's feelings correctly. *ScenarioAction* tests NPC's ability to correctly match the conversation with its facial expressions and body gestures. Our hypothesis is that low scores in *ScenarioEmotion* and *ScenarioAction* may be improved by changing the prompt but this requires further investigation. *ScenarioNoMemLim* consistently returns high APSAM scores. This is the scenario in which the player interacts with a single NPC asking questions related to the observations within the memory limit but asking for additional details. Players in this scenario express a clear intent by asking for additional details. The system can identify the specific information the player is seeking, making it easier to retrieve relevant details and generate additional responses.

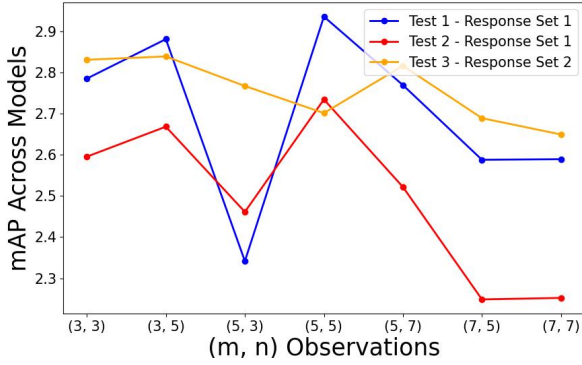
In human evaluation, we also noticed occurrences of plausible responses (ranging between 74% on the low end, and 97.2% on the high end - more details can be found in Table 1). The implausible responses are often repeating and we were able to identify patterns including: Lack of Contextual Understanding, Text-to-Speech Transcription Issues that contribute to incorrect responses, Lack of Appropriate Emotional Response, Misunderstandings of Possessive Pronouns (e.g., my, yours, theirs), LLM Hallucinations, Memory Loss and Lack of Understanding Gender.

Limitations A notable limitation pertains to the retrieval function module in our system. Regardless of the specific settings for the parameters 'm' and 'n', the retrieval process from the Observation Database is not entirely precise. This imprecision often results in the inclusion of irrelevant or incorrect observations. However, an advantage of our system is the integration with ChatGPT to generate responses. Those AI models are proficient in identifying and disregarding such noise, thereby ensuring the generation of responses that are both relevant and logical.

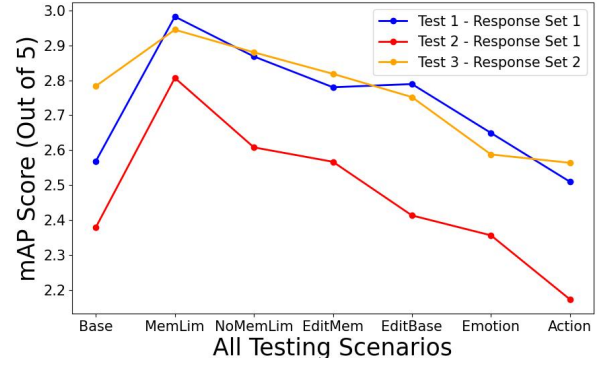
5 CONCLUSION & FUTURE WORK

In this paper, we present the design of the LLM-based agent system built into non-playable characters in VRChat platform. We evaluated the system on seven different Test Scenarios that helped us determine the most optimal parameters to achieve the highest Mean Average Precision of the Responses. We discussed the preliminary results, limitations and provided future research directions.

In the presented evaluation, we focused on determining the most optimal set of parameters to achieve the highest average precision in the NPC responses (generated by GPT-4). One critical part of the improvement of the AI agent's performance is conducting player evaluation as the next step. While our system is functioning as expected and supports interactions between one player and one NPC at a time, many factors contribute to the player experience and achieving seamless interaction. At present, our NPC response



(a) LLM-Judge mAP Score for NPC Response by Parameters



(b) LLM-Judge mAP Score for NPC Response by Scenarios

Figure 3: LLM-judge Evaluation Results

	$m = 3, n = 3$	$m = 3, n = 5$	$m = 3, n = 7$	$m = 5, n = 3$	$m = 5, n = 5$	$m = 5, n = 7$	$m = 7, n = 5$	$m = 7, n = 3$
Test 1	90%	93%	87.1%	74%	97.2%	90%	81.1%	79%
Test 2	87.5%	95.7%	92%	82%	91%	88%	96%	86%
Test 3	90%	95%	93%	93%	91.8%	88.8%	95%	86%
Mean Value	89.2%	94.57%	90.7%	83%	93.33%	88.93%	90.7%	83.7%

Table 1: NPC Plausible Response Rates Across Different Parameters Judged by Human Across All Tests

time is longer than human response time, and we have yet to assess the overall performance of our system from the player’s perspective. This will also entail looking at other present bottlenecks that contribute to prolonged response time from speech-to-text APIs to changing prompts and limiting response length as possible avenues to explore and improve the system performance.

To further assess the robustness and reliability of the AI models, conducting additional rounds of tests is necessary. This involves running the same GPT responses for more iterations and conducting test runs with different GPT responses to explore variations in responses. We also plans to recruiting more human evaluators for testing the system in future. We propose to explore fine-tuning hyper-parameters such as recency and relevancy. We also propose exploring the capability of generating advanced, abstract observations based on current low-level observations [29]. Additionally, we are aiming at utilizing LLM judge to mitigate the issue of “deep fake” occurrences in observations before adding observations into running memory [29, 40].

In addition, we see immense potential for various applications of our system, from practicing interviews in VR, planning and coordinating across multiple players, to expanding the system to include multiple NPCs and players simultaneously. This also deems consideration regarding the ethics, and exploring best practices for disclosing information about the nature of interactions in VR without breaking immersion during the gameplay.

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